

一切都是压缩
Everything Is Compression

作者说明

Author's Note

我叫谢凯凡。
My name is Kaifan Xie.

我是一名中文母语者，同时也是多语言及跨学科的长期爱好者。
I am a native Chinese speaker and a long-term enthusiast of multilingual and cross-disciplinary inquiry.

我对语言、认知、模型与解释机制的兴趣，并非来自单一学科，而是源于在不同语言与知识体系之间反复切换时，对压缩、失真与对齐问题的持续观察。
My interest in language, cognition, models, and explanation does not come from a single discipline, but from repeated observation of compression, distortion, and alignment while moving between languages and knowledge systems.

这篇文章并非试图给出终极答案，而是记录一种机制层视角：当我们解释世界时，究竟在做什么，又不可避免地舍弃了什么。
This essay does not aim to provide final answers, but to document a mechanism-level perspective: what we are actually doing when we explain the world—and what we inevitably discard.

Acknowledgements

感谢前面所有促成这篇文字的人。
Thanks to everyone who made this piece possible.

无论是直接的讨论、间接的启发，还是那些未被点名却真实存在的影响，它们都共同构成了这篇文章能够出现的背景条件。
Whether through direct discussion, indirect inspiration, or unnamed but real influences, all of them formed the conditions that allowed this text to exist.

这不是个人灵感的产物，而是一次在语言、经验与思考路径中逐步对齐的结果。
This is not the product of individual inspiration, but the result of gradual alignment across language, experience, and thought.

在此一并致谢。
Acknowledged with gratitude.

(马勒戈壁在这里老子忍不住吐槽一句老子差点被折磨疯了靠。)

世界本身不可直接处理。
The world itself cannot be processed directly.

它是连续的、高维的、同时发生的、充满噪声的。任何试图“如实理解世界”的行为，在机制上都是不可能的。

It is continuous, high-dimensional, simultaneous, and noisy. Any attempt to “understand the world as it is” is mechanically impossible.

人类之所以还能思考，只因为压缩。
Humans can think only because of compression.

感知是第一次压缩。
Perception is the first compression.

眼睛不是摄像机，耳朵不是录音机。你看到的不是光的连续分布，而是边缘、轮廓、物体；你听到的不是频谱，而是音素、节奏、意义。
Eyes are not cameras; ears are not recorders. You do not see continuous light, but edges, contours, objects; you do not hear spectra, but phonemes, rhythm, meaning.

大量信息在进入意识之前就被丢弃了，而且是结构性地丢弃。
Most information is discarded before reaching consciousness—and it is discarded structurally.

概念是第二次压缩。
Concepts are the second compression.

无数形态各异的具体事物，被压成一个词；无数事件与因果，被压成一个判断。
Countless distinct instances are compressed into a single word; countless events and causes into a single judgement.

概念的本质不是总结，而是忽略差异。
The essence of a concept is not summarisation, but the deliberate ignoring of differences.

如果不忽略，你就无法复用。
Without ignoring differences, reuse is impossible.

语言是第三次压缩。
Language is the third compression.

语言不是为精确而生，而是为共享而生。
Language is not born for precision, but for sharing.

它必须在不同个体之间可对齐、可传播、可快速调用，因此必然牺牲细节。
It must align across individuals, propagate efficiently, and be rapidly callable—so detail must be sacrificed.

语言不是现实的镜子，而是群体协作所需的最低带宽协议。
Language is not a mirror of reality, but a low-bandwidth protocol for collective coordination.

解释，是在已有压缩之上的再压缩。
Explanation is compression built on top of existing compression.

一篇文章、一个理论、一次说明，本质都是把已经被压过世界，再压一次，使其能被他人接收。
An article, a theory, an explanation—all compress an already-compressed world once more so it can be received by others.

所谓“讲清楚”，不是增加信息，而是减少自由度。
To “explain clearly” is not to add information, but to reduce degrees of freedom.

模型也是压缩。

Models are also compression.

无论是数学模型、理论框架，还是你脑中的直觉判断，都是为了在有限算力下保留关键结构、舍弃其余部分。

Whether mathematical models, theoretical frameworks, or intuition, all exist to preserve key structure under limited computational capacity.

模型的价值不在“是否真实”，而在“在某个尺度下是否可用”。

A model's value lies not in being “true”, but in being usable at a given scale.

理解不是接近真相，而是获得可复用的压缩。

Understanding is not approaching truth, but acquiring reusable compression.

你真正理解某件事，不是因为你掌握了所有细节，而是因为你能在不同情境中，用同一套压缩结构做出不崩溃的判断。

You truly understand something when the same compressed structure holds across contexts without collapsing.

因此，所有解释都是有偏的。

Therefore, all explanations are biased.

没有“完全正确的说法”，只有在特定目的、特定尺度、特定边界内不失效的压缩。

There is no perfectly correct account—only compression that does not fail within specific purposes, scales, and boundaries.

语言失败，并不是语言不好，而是压缩失配。

When language fails, it is not because language is bad, but because compression is mismatched.

当问题复杂度超过当前压缩层级，语言就开始制造幻觉：看似清楚，实则错位。

When problem complexity exceeds the compression level, language generates illusions—seeming clear while being misaligned.

补充：中文中的「压缩」与「解释」

Addendum: “Compression” and “Explanation” in Chinese

中文里，“解释”这个词本身就暴露了机制。

In Chinese, the word “解释” already exposes the mechanism.

解，是拆解结构；

“解” means to break apart structure;

释，是重新释放为可理解形式。

“释” means to release it again in an understandable form.

解释并不是把世界原样展开，而是：

Explanation is not unfolding the world as-is, but:

拆掉细节，保留骨架，再用更少的符号重新封装。

removing detail, preserving skeletons, and repackaging with fewer symbols.

中文是一种天然的高压缩语言。

Chinese is a naturally high-compression language.

一个词、一个字，往往压缩了大量历史、关系与语境。

A single character or word often compresses history, relations, and context.

这使得中文使用者长期在“极端压缩块”中思考，只是大多数人并不自觉。
This forces Chinese speakers to think in extreme compression blocks—usually without awareness.

当你开始意识到“解释本身就是压缩”，并不是灵感闪现。
When you realise that explanation itself is compression, it is not a sudden insight.

那只是你终于看见了自己一直在使用的操作。
It is simply recognising the operation you have always been using.

认知的全部历史，就是压缩策略的演化史。
The entire history of cognition is the evolution of compression strategies.

从感知到概念，从语言到理论，从个人理解到群体共识，没有一层不是在做同一件事。
From perception to concepts, language to theory, individual understanding to collective consensus—
every layer does the same thing.

在不可承受的复杂性面前，构造一个还能用的世界。
Faced with unbearable complexity, we construct a world that still works.

所以你突然意识到“好像所有东西都是这个”，并不夸张。
So when you realise “it all seems to be this,” it is not exaggeration.

那是你看见了机制层。
You are seeing the mechanism layer.

世界没有变简单。
The world did not become simpler.

只是你终于看见了，人类一直靠什么活下来。
You finally saw what humans have always relied on to survive.

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The views expressed focus on structural issues of language, cognition, and explanation, and do not
constitute academic conclusions, professional advice, or authoritative positions.

由于“解释”本身即是一种压缩，
本文不可避免地省略细节、简化复杂性，并在特定尺度下成立。
Because explanation itself is a form of compression, this text inevitably omits details, simplifies
complexity, and holds only within specific scales.

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